Day-4

GPU and cooling systems theory.

-Basics of GPUs and cooling systems (Air vs Liquid cooling)

-Identify a GPU and cooling systems in a pc setup.

Q-1 :-Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated .

Ans- Photo of GPU Geforce GTX 1650 from NVIDEA.



GPU Specifications :-

Brand	:	NVIDIA.
Model number	:	GeForce GTX 1650.
GPU type	:	Dedicated.
Memory	:	4 GB GDDR5 or GDDR6, depending on model.
Main use	:	Gaming and video editing.
Function	:	Displays graphics on the screen.

-A GPU (Graphics Processing Unit) processes images, videos, and game graphics. A dedicated GPU is a separate graphics card that usually provides better gaming performance than basic integrated graphics.

Note: The GTX 1650 is only an example, not a confirmed model in your computer. To identify your actual GPU, press Ctrl + Shift + Esc → Performance → GPU and write down the model name displayed.

Q-2 :- Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

Ans-Cooling system photo and other details is as under.



A cooling system keeps a computer from getting too hot. It uses fans, metal heatsinks, or liquid to remove heat from the CPU and other components.

Main cooling components :-

Component	**:**	**Function.**
Cooling fan	**:**	**Moves air.**
Heatsink	**:**	**Absorbs heat.**
Radiator	**:**	**Cools liquid.**
Pump	**:**	**Moves liquid.**
Tubes	**:**	**Carry liquid.**

Short assignment answer: The cooling system removes heat from the computer. Fans move air, the heatsink absorbs heat, and liquid cooling uses a pump, tubes, and radiator to carry heat away.

Q-3 :- Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Ans-Air Cooling vs Liquid Cooling for GPU are as under.



-Air cooling: Uses fans and a metal heatsink to remove heat from the GPU.

-Liquid cooling: Uses liquid, tubes, a pump, and a radiator to remove heat from the GPU.

Conclusion: Air cooling is cheaper and easier to maintain. Liquid cooling can provide excellent cooling but usually costs more and is more complex to install.

Q-4 :-Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

Ans- I would choose air cooling for my gaming PC because it is affordable, easy to install, and simple to maintain. It uses fans and a heatsink to remove heat from the GPU while playing graphics-heavy games and streaming IPL matches. A good air cooler can keep the GPU at a safe temperature during normal gaming. Liquid cooling can provide better cooling for some high-performance GPUs, but it costs more and needs extra care.

Note- My personal opinion is if you have small size budget then go for Air cooling GPU and if you have no budget restriction and you are performing 6K,8K (video formate) resolution rendering ,editing or you want to require peak performance for your professional work so you can go for Liquid cooling GPU.